AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

APRIL/ MAY 2018

Biology Paper II

Time: 2 hours 25 minutes Marks: 40

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INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are NINE questions. Answer ALL questions. Questions 8 and 9 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

Q.1. (Total 5 Marks)

a. Identify the branch of science that shows relationship with biology in the given examples. (2 Marks)

S. No.	Example	Branch of Science
1	Students of grade IX measured the heights of 50 individuals in their class and represented the data in the form of a bar graph.	
2	In Pakistan, the total value of fish export for the years 2014 and 2015 was US$ 253.497 million.	

b. State any THREE salient features of prokaryotes. (3 Marks)

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Q.2. (Total 4 Marks)

Fahad conducts an experiment to investigate the effect of humidity on the rate of transpiration in plants. After a week, he records the following data.

Day	Humidity (%)	Transpiration Rate ($g/m^2/hr$)
1	40	198
2	61	188
3	62	186
4	75	150
5	78	80
6	80	60
7	88	57

a. Present the given data in the form of a graph. (3 Marks)

Space for graph

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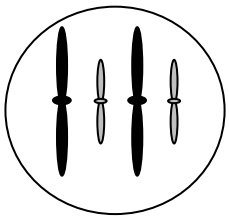
b. What result can be concluded from the given data? (1 Mark)

_____

_____

Q.3. (Total 3 Marks)

The given diagram shows a cell with four chromosomes.



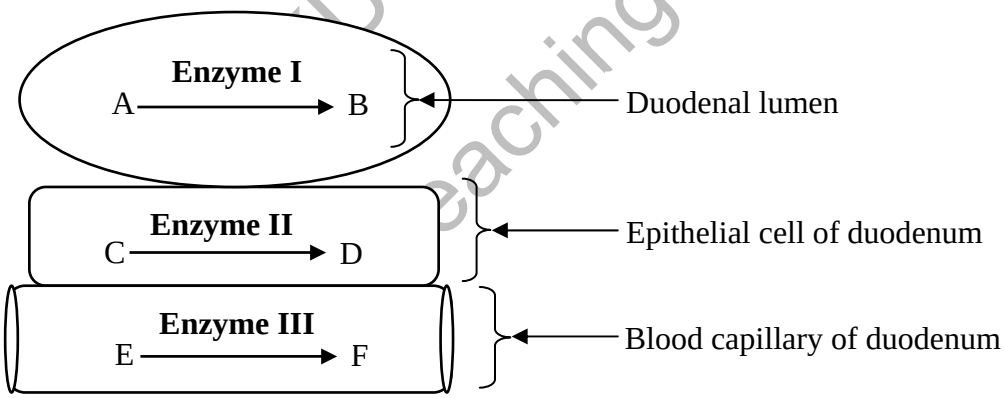
Draw the condition of chromosomes of the given cell during **prophase I, metaphase I and metaphase II** when this cell undergoes meiosis.

Prophase I	Metaphase I	Metaphase II

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Q.4. (Total 2 Marks)

The given diagram illustrates the activity of three enzymes in the duodenum of human beings.



Classify the enzymes, **I**, **II** and **III**, as intracellular or extracellular.

Intracellular Enzyme	Extracellular Enzyme

Q.5. (Total 3 Marks)

List any THREE factors that affect the rate of photosynthesis in plants.

Q.6. (Total 3 Marks)

Mention any THREE protective functions of blood against diseases in the human body.

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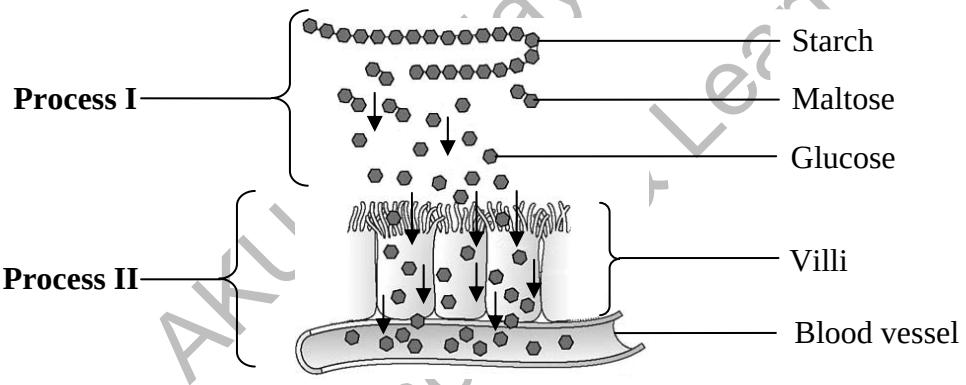
Q.7. (Total 5 Marks)

a. The given table shows the percentage of nutrients present in two food items.

Food Item	Protein (%)	Fat (%)	Fibre (%)
I	13	11	24
II	7	18	6

Which of the food items, **I** or **II**, would provide more energy? Support your answer with a suitable reason. (2 Marks)

b. The given diagram illustrates two processes occurring in the small intestine of human beings.



i. Identify the processes, **I** and **II**. (2 Marks)

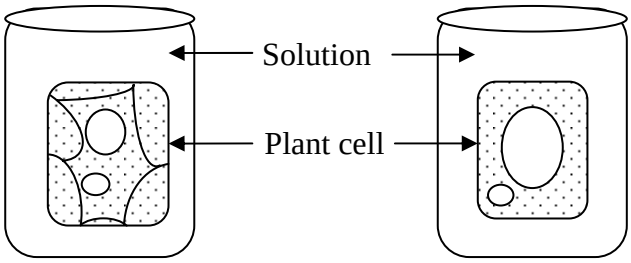
ii. What is the significance of villi for process **II**? (1 Mark)

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Q.8. (Total 7 Marks)

EITHER

a. The given diagrams, **I** and **II**, show the effect of different solutions on plant cells.



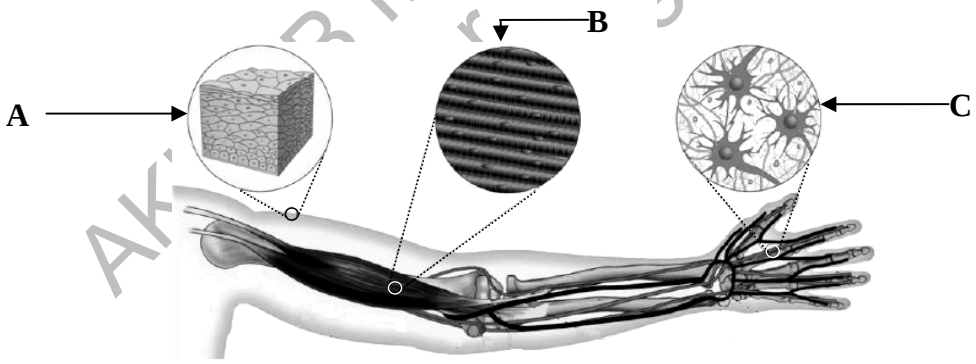
I **II**

i. What is meant by tonicity? (1 Mark)

ii. Explain the effect of the solution on the cell in each case with reference to osmosis. (6 Marks)

OR

b. The given diagram shows different types of tissues present in the human arm.



i. State any ONE feature of animal tissues. (1 Mark)

ii. Describe the specificities and functions of tissues, labelled as **A**, **B** and **C** shown in the given diagram. (6 Marks)

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Q.9. (Total 8 Marks)

EITHER

a. Explain the process of oxidation of glucose that takes place in the cells of human beings.

(**Hint:** Refer to the mechanism of cellular respiration.)

OR

b. The human heart is adapted to its function.

Explain the given statement with reference to the structure of any FOUR parts of the heart and their relevant functions.

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